MSG No. 123.349 P_C4/mmm [Type IV, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `4/mmm`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32]`

Table 2: Wyckoff site: `2b`, site symmetry: `4/mmm`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32]`

Table 3: Wyckoff site: `2c`, site symmetry: `4'/mmm'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,4,5,6,9,12,13,14,18,19,23,24,26,27,31,32]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,3,7,8,10,11,15,16,17,20,21,22,25,28,29,30]`

Table 4: Wyckoff site: `2d`, site symmetry: `4'/mmm'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,4,5,6,9,12,13,14,18,19,23,24,26,27,31,32]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,3,7,8,10,11,15,16,17,20,21,22,25,28,29,30]`

Table 5: Wyckoff site: `4e`, site symmetry: `m.mm'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[1,8,14,15,22,23,25,32]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[2,4,11,13,19,21,26,28]`
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[3,5,10,12,18,20,27,29]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[6,7,9,16,17,24,30,31]`

Table 6: Wyckoff site: `4f`, site symmetry: `m.mm'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,8,14,15,22,23,25,32]`
2	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[2,4,11,13,19,21,26,28]`
3	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[3,5,10,12,18,20,27,29]`
4	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[6,7,9,16,17,24,30,31]`

Table 7: Wyckoff site: `4g`, site symmetry: `4mm`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,6,12,13,15,16]`
2	$[0,\ 0,\ -z]$	`[4,5,7,8,9,10,11,14]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[17,18,19,22,28,29,31,32]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[20,21,23,24,25,26,27,30]`

Table 8: Wyckoff site: `4h`, site symmetry: `4'mm'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,6,12,13,18,19,31,32]`
2	$\left[\frac{1}{2},\ 0,\ z\right]$	`[2,3,15,16,17,22,28,29]`
3	$\left[0,\ \frac{1}{2},\ -z\right]$	`[4,5,9,14,23,24,26,27]`
4	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[7,8,10,11,20,21,25,30]`

Table 9: Wyckoff site: `8i`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[1,15,22,32]`
2	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	`[2,13,19,28]`
3	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	`[3,12,18,29]`
4	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	`[4,11,21,26]`
5	$\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$	`[5,10,20,27]`
6	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[6,16,17,31]`
7	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	`[7,9,24,30]`
8	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	`[8,14,23,25]`

Table 10: Wyckoff site: `8j`, site symmetry: `mm2.`

No.	position	mapping
1	$[0,\ y,\ 0]$	`[1,5,12,14]`
2	$[-y,\ 0,\ 0]$	`[2,8,11,15]`
3	$[y,\ 0,\ 0]$	`[3,7,10,16]`
4	$[0,\ -y,\ 0]$	`[4,6,9,13]`
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$	`[17,21,28,30]`
6	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ 0\right]$	`[18,24,27,31]`
7	$\left[y+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[19,23,26,32]`
8	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	`[20,22,25,29]`

Table 11: Wyckoff site: `8k`, site symmetry: `mm2.`

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{2}\right]$	`[1,5,12,14]`
2	$\left[-y,\ 0,\ \frac{1}{2}\right]$	`[2,8,11,15]`

continued ...

Table 11

No.	position	mapping
3	$\left[y,\, 0,\, \frac{1}{2}\right]$	[3,7,10,16]
4	$\left[0,\, -y,\, \frac{1}{2}\right]$	[4,6,9,13]
5	$\left[\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[17,21,28,30]
6	$\left[\frac{1}{2}-y,\, \frac{1}{2},\, \frac{1}{2}\right]$	[18,24,27,31]
7	$\left[y+\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}\right]$	[19,23,26,32]
8	$\left[\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	[20,22,25,29]

Table 12: Wyckoff site: 8l, site symmetry: m.2m

No.	position	mapping
1	$[x,\, x,\, 0]$	[1,7,14,16]
2	$[-x,\, x,\, 0]$	[2,5,11,12]
3	$[x,\, -x,\, 0]$	[3,4,10,13]
4	$[-x,\, -x,\, 0]$	[6,8,9,15]
5	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, 0\right]$	[17,23,30,32]
6	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, 0\right]$	[18,21,27,28]
7	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, 0\right]$	[19,20,26,29]
8	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, 0\right]$	[22,24,25,31]

Table 13: Wyckoff site: 8m, site symmetry: m.2m

No.	position	mapping
1	$\left[x,\, x,\, \frac{1}{2}\right]$	[1,7,14,16]
2	$\left[-x,\, x,\, \frac{1}{2}\right]$	[2,5,11,12]
3	$\left[x,\, -x,\, \frac{1}{2}\right]$	[3,4,10,13]
4	$\left[-x,\, -x,\, \frac{1}{2}\right]$	[6,8,9,15]
5	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[17,23,30,32]
6	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[18,21,27,28]
7	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[19,20,26,29]
8	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[22,24,25,31]

Table 14: Wyckoff site: 8n, site symmetry: m.2'm'

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, 0\right]$	[1,14,23,32]
2	$\left[\frac{1}{2}-x,\, x,\, 0\right]$	[2,11,21,28]
3	$\left[x+\frac{1}{2},\, -x,\, 0\right]$	[3,10,20,29]
4	$\left[x,\, \frac{1}{2}-x,\, 0\right]$	[4,13,19,26]
5	$\left[-x,\, x+\frac{1}{2},\, 0\right]$	[5,12,18,27]
6	$\left[-x,\, \frac{1}{2}-x,\, 0\right]$	[6,9,24,31]
7	$\left[x+\frac{1}{2},\, x,\, 0\right]$	[7,16,17,30]
8	$\left[\frac{1}{2}-x,\, -x,\, 0\right]$	[8,15,22,25]

Table 15: Wyckoff site: `8o`, site symmetry: `m.2'm'`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	`[1,14,23,32]`
2	$\left[\frac{1}{2}-x,\, x,\, \frac{1}{2}\right]$	`[2,11,21,28]`
3	$\left[x+\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	`[3,10,20,29]`
4	$\left[x,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	`[4,13,19,26]`
5	$\left[-x,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	`[5,12,18,27]`
6	$\left[-x,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	`[6,9,24,31]`
7	$\left[x+\frac{1}{2},\, x,\, \frac{1}{2}\right]$	`[7,16,17,30]`
8	$\left[\frac{1}{2}-x,\, -x,\, \frac{1}{2}\right]$	`[8,15,22,25]`

Table 16: Wyckoff site: `16p`, site symmetry: `m..`

No.	position	mapping
1	$[x,\, y,\, 0]$	`[1,14]`
2	$[-y,\, x,\, 0]$	`[2,11]`
3	$[y,\, -x,\, 0]$	`[3,10]`
4	$[x,\, -y,\, 0]$	`[4,13]`
5	$[-x,\, y,\, 0]$	`[5,12]`
6	$[-x,\, -y,\, 0]$	`[6,9]`
7	$[y,\, x,\, 0]$	`[7,16]`
8	$[-y,\, -x,\, 0]$	`[8,15]`
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, 0\right]$	`[17,30]`
10	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, 0\right]$	`[18,27]`
11	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, 0\right]$	`[19,26]`
12	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, 0\right]$	`[20,29]`
13	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, 0\right]$	`[21,28]`
14	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, 0\right]$	`[22,25]`
15	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, 0\right]$	`[23,32]`
16	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, 0\right]$	`[24,31]`

Table 17: Wyckoff site: `16q`, site symmetry: `m..`

No.	position	mapping
1	$\left[x,\, y,\, \frac{1}{2}\right]$	`[1,14]`
2	$\left[-y,\, x,\, \frac{1}{2}\right]$	`[2,11]`
3	$\left[y,\, -x,\, \frac{1}{2}\right]$	`[3,10]`
4	$\left[x,\, -y,\, \frac{1}{2}\right]$	`[4,13]`
5	$\left[-x,\, y,\, \frac{1}{2}\right]$	`[5,12]`
6	$\left[-x,\, -y,\, \frac{1}{2}\right]$	`[6,9]`
7	$\left[y,\, x,\, \frac{1}{2}\right]$	`[7,16]`
8	$\left[-y,\, -x,\, \frac{1}{2}\right]$	`[8,15]`
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}\right]$	`[17,30]`
10	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	`[18,27]`
11	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	`[19,26]`
12	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	`[20,29]`

continued ...

Table 17

No.	position	mapping
13	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[21,28]
14	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	[22,25]
15	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[23,32]
16	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[24,31]

Table 18: Wyckoff site: 16r, site symmetry: .m.

No.	position	mapping
1	$[0,\, y,\, z]$	[1,12]
2	$[-y,\, 0,\, z]$	[2,15]
3	$[y,\, 0,\, z]$	[3,16]
4	$[0,\, -y,\, -z]$	[4,9]
5	$[0,\, y,\, -z]$	[5,14]
6	$[0,\, -y,\, z]$	[6,13]
7	$[y,\, 0,\, -z]$	[7,10]
8	$[-y,\, 0,\, -z]$	[8,11]
9	$\left[\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[17,28]
10	$\left[\frac{1}{2}-y,\, \frac{1}{2},\, z\right]$	[18,31]
11	$\left[y+\frac{1}{2},\, \frac{1}{2},\, z\right]$	[19,32]
12	$\left[\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[20,25]
13	$\left[\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	[21,30]
14	$\left[\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$	[22,29]
15	$\left[y+\frac{1}{2},\, \frac{1}{2},\, -z\right]$	[23,26]
16	$\left[\frac{1}{2}-y,\, \frac{1}{2},\, -z\right]$	[24,27]

Table 19: Wyckoff site: 16s, site symmetry: ..m

No.	position	mapping
1	$[x,\, x,\, z]$	[1,16]
2	$[-x,\, x,\, z]$	[2,12]
3	$[x,\, -x,\, z]$	[3,13]
4	$[x,\, -x,\, -z]$	[4,10]
5	$[-x,\, x,\, -z]$	[5,11]
6	$[-x,\, -x,\, z]$	[6,15]
7	$[x,\, x,\, -z]$	[7,14]
8	$[-x,\, -x,\, -z]$	[8,9]
9	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, z\right]$	[17,32]
10	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, z\right]$	[18,28]
11	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, z\right]$	[19,29]
12	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, -z\right]$	[20,26]
13	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, -z\right]$	[21,27]
14	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, z\right]$	[22,31]
15	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	[23,30]
16	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, -z\right]$	[24,25]

Table 20: Wyckoff site: `16t`, site symmetry: `..m'`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, z\right]$	`[1,32]`
2	$\left[\frac{1}{2}-x,\, x,\, z\right]$	`[2,28]`
3	$\left[x+\frac{1}{2},\, -x,\, z\right]$	`[3,29]`
4	$\left[x,\, \frac{1}{2}-x,\, -z\right]$	`[4,26]`
5	$\left[-x,\, x+\frac{1}{2},\, -z\right]$	`[5,27]`
6	$\left[-x,\, \frac{1}{2}-x,\, z\right]$	`[6,31]`
7	$\left[x+\frac{1}{2},\, x,\, -z\right]$	`[7,30]`
8	$\left[\frac{1}{2}-x,\, -x,\, -z\right]$	`[8,25]`
9	$\left[-x,\, \frac{1}{2}-x,\, -z\right]$	`[9,24]`
10	$\left[x+\frac{1}{2},\, -x,\, -z\right]$	`[10,20]`
11	$\left[\frac{1}{2}-x,\, x,\, -z\right]$	`[11,21]`
12	$\left[-x,\, x+\frac{1}{2},\, z\right]$	`[12,18]`
13	$\left[x,\, \frac{1}{2}-x,\, z\right]$	`[13,19]`
14	$\left[x,\, x+\frac{1}{2},\, -z\right]$	`[14,23]`
15	$\left[\frac{1}{2}-x,\, -x,\, z\right]$	`[15,22]`
16	$\left[x+\frac{1}{2},\, x,\, z\right]$	`[16,17]`

Table 21: Wyckoff site: `32u`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$[-y,\, x,\, z]$	`[2]`
3	$[y,\, -x,\, z]$	`[3]`
4	$[x,\, -y,\, -z]$	`[4]`
5	$[-x,\, y,\, -z]$	`[5]`
6	$[-x,\, -y,\, z]$	`[6]`
7	$[y,\, x,\, -z]$	`[7]`
8	$[-y,\, -x,\, -z]$	`[8]`
9	$[-x,\, -y,\, -z]$	`[9]`
10	$[y,\, -x,\, -z]$	`[10]`
11	$[-y,\, x,\, -z]$	`[11]`
12	$[-x,\, y,\, z]$	`[12]`
13	$[x,\, -y,\, z]$	`[13]`
14	$[x,\, y,\, -z]$	`[14]`
15	$[-y,\, -x,\, z]$	`[15]`
16	$[y,\, x,\, z]$	`[16]`
17	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	`[17]`
18	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z\right]$	`[18]`
19	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z\right]$	`[19]`
20	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	`[20]`
21	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$	`[21]`
22	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	`[22]`
23	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	`[23]`
24	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, -z\right]$	`[24]`
25	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, -z\right]$	`[25]`

continued ...

Table 21

No.	position	mapping
26	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ -z\right]$	[26]
27	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ -z\right]$	[27]
28	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	[28]
29	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	[29]
30	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[30]
31	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z\right]$	[31]
32	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z\right]$	[32]